

Question 3

The World weather service has collected data on the number of Typhoons to hit the island of "Island X" since 1994 and has observed the following numbers by year

TABLE 1 – NUMBER OF TYPHOONS TO HIT "ISLAND X" SINCE 1994

YEAR	NUMBER OF TYPHOONS
1994	0
1995	8
1996	5
1997	4
1998	2
1999	1
2000	3
2001	2
2002	6
2003	3
2004	3

A. Assume that the Number of Typhoons hitting "Island X" is a Poisson Process then

- Derive a point estimate for the expected number of Typhoons to hit "Island X" in a year
- Give a 95% confidence interval for the Expected Number of Typhoons to hit "Island X" in a year

B. Now assume that the Number of Typhoons hitting "Island X" is best Approximated with a Normal Distributed then

- Derive a point estimate for the expected number of Typhoons to hit "Island X" in a year
- Give a 95% confidence interval for the Expected Number of Typhoons to hit "Island X" in a year

In 2002 the World Weather service started recording size (i.e. Category) of typhoon and observed that 6 of the 12 typhoons occurring in 2002, 2003 and 2004 were greater than category 2. Assume that category of Typhoon is independent of number of Typhoons that are hitting during the year.

C. Based on this information and Table 1 up above, what is the expected number of Typhoons > Category 2 to hit "Island X" in a year.